

NIRAV SURABHI

Half on a **screen**, half on a **workbench** — software + mechanical engineering.

Brambleton, VA niravsurabhi@gmail.com niravsurabhi.com linkedin.com/in/niravss
github.com/21J3phy

EDUCATION

Purdue University

FALL 2026 (INCOMING)

[B.S. Mechanical Engineering](#)

Independence High School

GRADUATING JUN 2026

[Brambleton, VA](#) · [Senior](#)

SOFTWARE — FOUNDER PROJECTS

Chad vs. Chud

2026–NOW · CO-FOUNDER & CEO

[chadvschud.com](#)

- Live web app where people join real-time sessions and a custom CNN ranks faces from live video.
- Trained the model, wrote the backend, shipped on Fly.io. **10,000+ users, 100+ concurrent.**

[Python](#) · [PyTorch](#) · [FastAPI](#) · [Next.js](#) · [Tailwind](#) · [Fly.io](#)

b2alpha.io

2026–NOW · CO-FOUNDER & CEO

[b2alpha.io](#)

- Open agent-to-agent protocol: secure messaging + DNS integration.
- Prompt engineering to make models emit structured output other agents can parse. **2,000+ business users.**

[TypeScript](#) · [Node.js](#) · [OpenAI API](#) · [Vercel](#) · [Supabase](#) · [DNS](#)

PredictionView

2026–NOW · CO-FOUNDER & CEO

[predictionview.app](#)

- Analytics for prediction markets — charts, signals, comparisons.
- Own the backend, tests, and most of the debugging.

[Next.js](#) · [TypeScript](#) · [Supabase](#) · [Recharts](#)

Khali

2025–2026 · CO-FOUNDER & CEO

[khali.info](#)

- News app with a short-form (Reels-style) video feed, built to get past the spin.
- Submitted to Y Combinator's Spring 2026 batch.

React Native · Expo · Supabase · News APIs

MECHANICAL / HARDWARE

Passive Vortex Propulsion

2024–2025 · FIRST AUTHOR

- Built a wind tunnel from scratch; soldered and calibrated the load cells with an Arduino; ran CFD in Fusion 360.
- First demonstration of passive vortex propulsion in air — published first-author paper; AIAA award & NASA Goddard invitation.

Arduino · C++ · load cells · Fusion 360 · CFD · Python

FTC Robotics — Absolute Zero (12096)

2022–2026 · CAPTAIN & LEAD PROGRAMMER

- Four years; CADded and built every part in Fusion 360.
- Wrote autonomous and tele-op in Java — PID loops, sensor fusion, odometry. Reached the Chesapeake championship.

Electric Vehicle & Salinity Sensor

SCIENCE OLYMPIAD

- Autonomous EV that accelerates then brakes to a target distance — Arduino, goBILDA motor, quadrature encoder, PID.
 - Arduino salt-water salinity sensor reading conductivity.
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EXPERIENCE

AI Engineer & SWE Intern

FEB 2021–NOW

Ember Learning (formerly DeAP Learning Labs)

- Wrote the LLM prompts behind AP-tutor chatbots; built automated AI grading for AP DBQs/FRQs against the official rubric.
- Indexed lecture transcripts in a vector DB for timestamp-level search; led the company website.
- Team landed a Microsoft & OpenAI grant and crossed a \$1M+ valuation.

CAD Intern

JUN 2025–MAY 2026

I3 Simulations

- 3D models and haptic-feedback prototypes for medical-training simulators.

Bike Mechanic

JUN 2021–AUG 2025

The Bike Lane · intern → part-time

- Built bikes from the frame up; ran final safety checks.

SKILLS		HONORS & MUSIC	
LANGUAGES	Java, Python, JavaScript, TypeScript, HTML, CSS, Git	▪ AIAA Aerospace Award (RSEF special honor → NASA Goddard invitation) ▪ Captain, Science Olympiad ▪ Presidential Volunteer Service Award (Silver) ▪ Central Virginia Governor's School for Science & Technology — Mathematics, 2024 ▪ 1st place, Agaudo guitar competition · All-Virginia Guitar Ensemble	
AI / ML	Prompt engineering, LLMs, CNNs, computer vision, agent architecture		
CLOUD	Fly.io, Vercel, Supabase, Fusion 360		
HARDWARE	Arduino, load cells, soldering, multimeter, PID control		